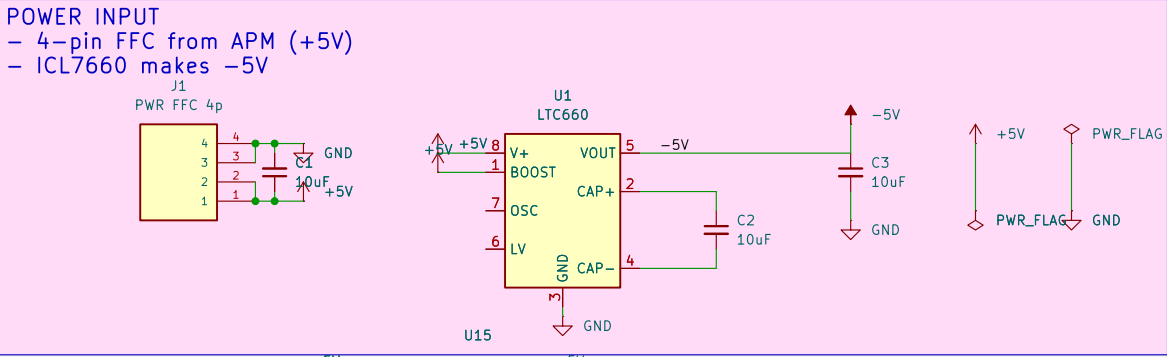
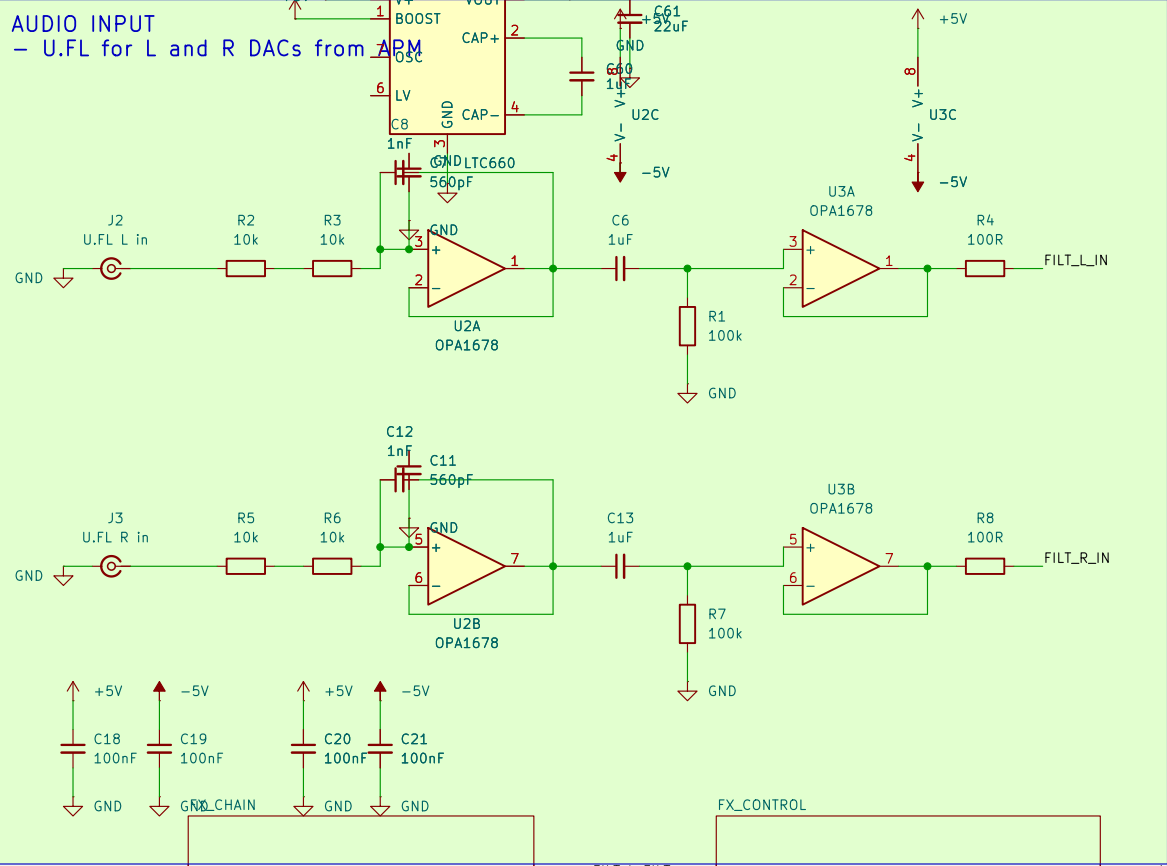
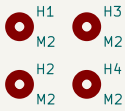


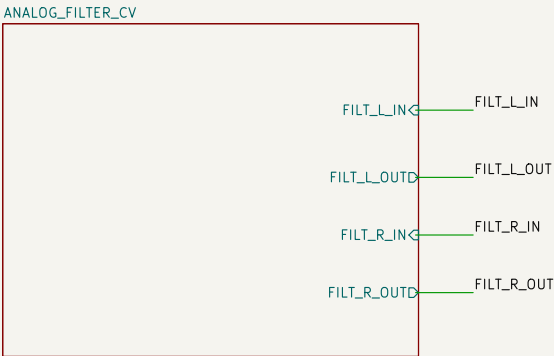
AUDIO BOARD – front end: power, U.FL audio in, reconstruction filter + buffers, 1/4in L/R + 3.5mm out



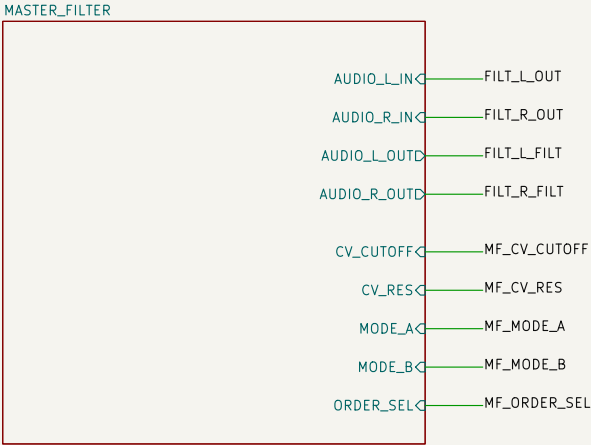
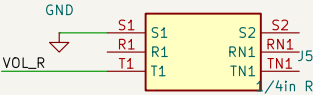
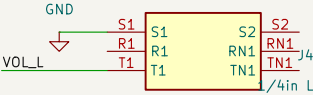
OUTPUTS – 1/4in L/R (ACJS-MHOM) + 3.5mm headphone/debug



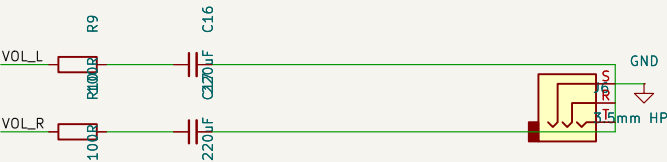
ANALOG PATH – each channel's buffer output (FILT_x_IN) goes into the analog sub-sheet; FILT_x_OUT returns to the jacks



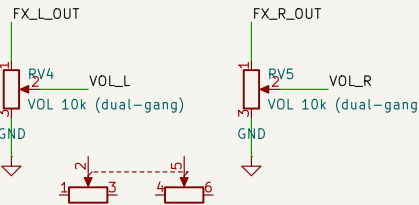
File: ANALOG_FILTER_CV.kicad_sch



File: MASTER_FILTER.kicad_sch

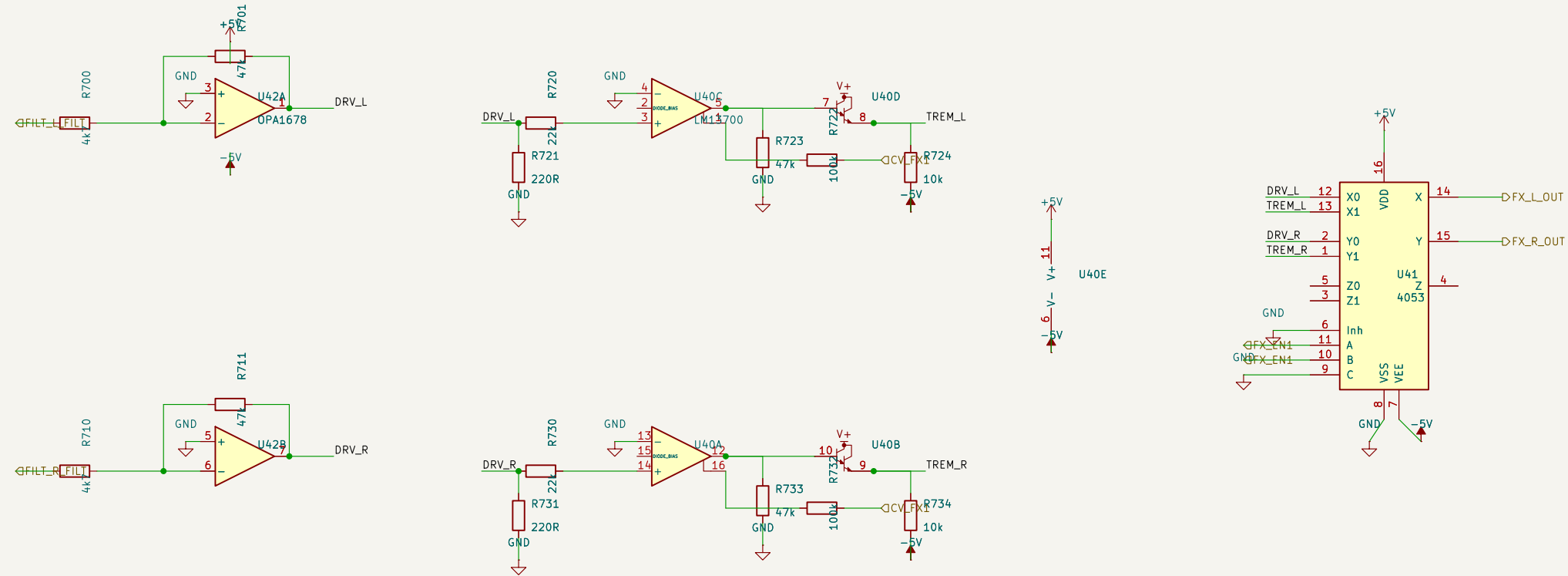


MASTER VOLUME – RV4/RV5 = one dual-gang pot (shared L/R knob); attenuates final output to the jacks/HP



RV6 R_Potentiometer_Dual		
Sheet: /		
File: AUDIO_BOARD.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 1/5

FX CHAIN - ¹DRIVE (soft clip) + ²TREMOLO (OTA VCA, LFO=CV_FX1); each bypassable (FX_EN). In: ³FILT_x_FILT (from SVF) -> ⁴Out: ⁵FX_x_OUT (to volume)



DECOUPLING 100nF/C



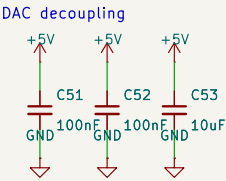
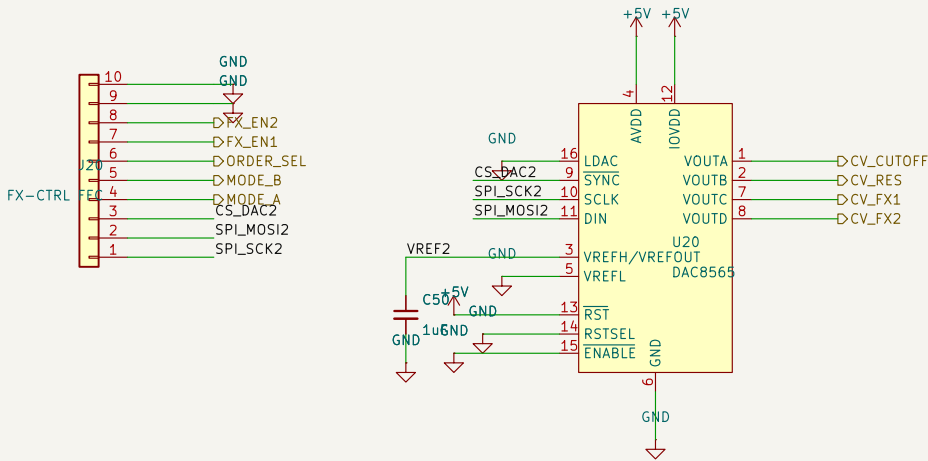
log FX ~ simulate/verify diode clip levels and OTA VCA vs LM13700 datasheet

NOTE: analog FX – simulate/verify diode clip levels and OTA VCA vs LM13700 datasheet before fab.

Sheet: /FX_CHAIN/		
File: FX_CHAIN.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 5/5

FX CONTROL – 2nd FX-control FFC (SPI + mode/order/enable) -> DAC8565 #2 makes CV_CUTOFF/CV_RES (+2 spare FX CVs); digital selects pass through

FX-CTRL FFC: 1SCK 2MOSI 3CS_DAC2 4MODE_A 5MODE_B 6ORDER_SEL 7FX_EN2 8FX_EN1 9VREF2 10GND



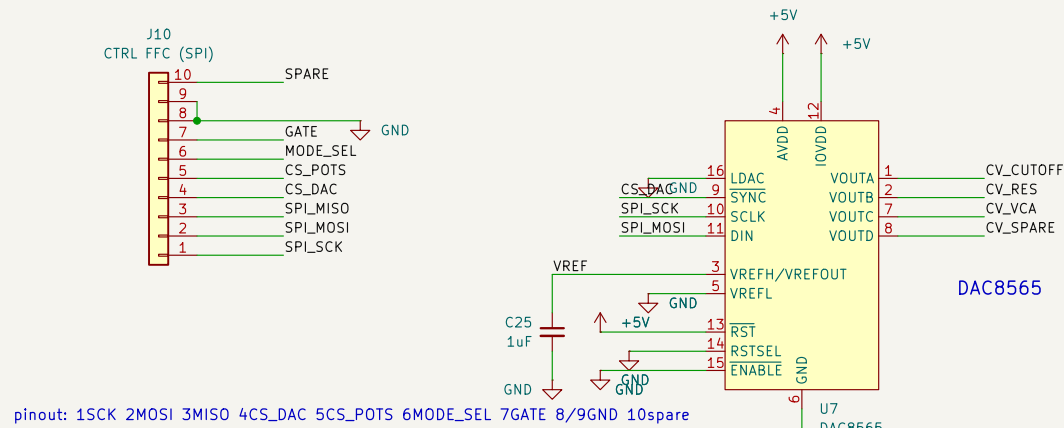
outputs to filter/FX: CV_CUTOFF, CV_RES (-> SVF), CV_FX1/CV_FX2 (-> future phaser/chorus/tremolo); MODE_A/B, ORDER_SEL, FX_EN1/2 digital.

Knobs/switches live on the controller board; STM32 reads them, drives this DAC over SPI and sets the digital lines.

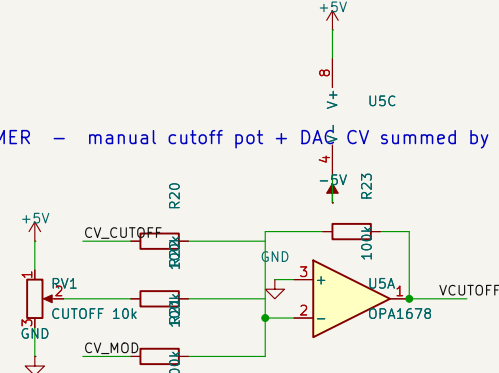
ANALOG FILTER / CV (stereo) - SPI in (10-pin FFC) -> DAC8565 makes CVs -> per-channel LM13700 VCF/VCA; CD4053 selects analog or FPGA-digital

CONTROL - 10-pin FFC (SPI from APM)

TOGGLE - CD4053: MODE_SEL selects analog (VCA_x_OUT) vs bypass (FILT_x_IN) for BOTH L and R

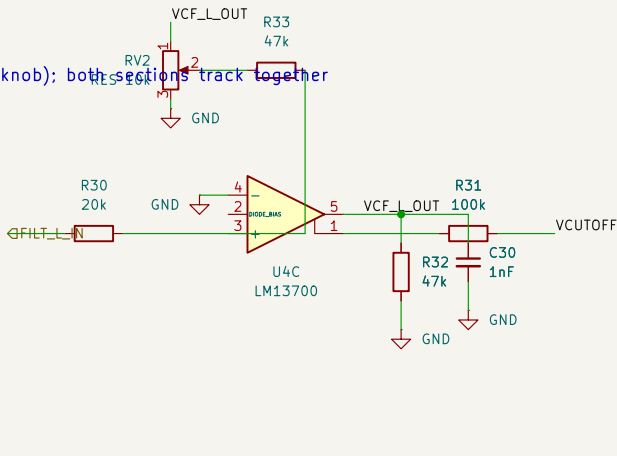


CV SUMMER - manual cutoff pot + DAC CV summed by U5 -> VCUTOFF (drives both L and R VCFs)

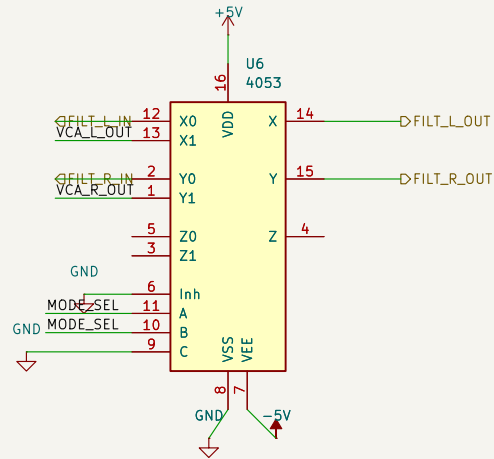
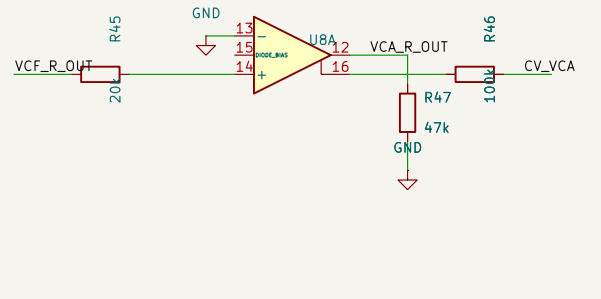
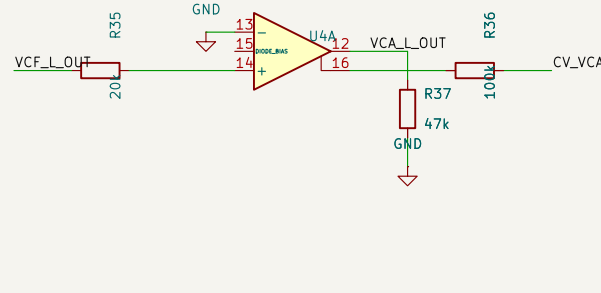
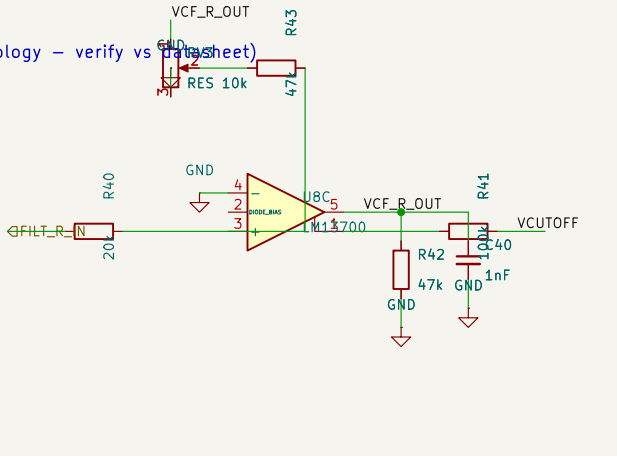


VCF+VCA L - LM13700 OTA lowpass then OTA VCA (NOTE: OTA/labc topology - verify vs datasheet)

RESONANCE - RV2 (L) + RV3 (R) = one dual-gang pot (shared stereo knob); both sections track together



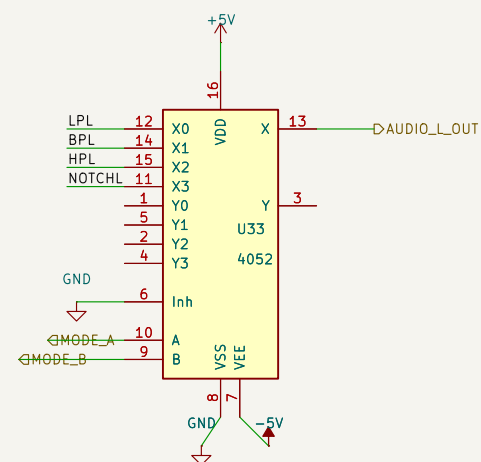
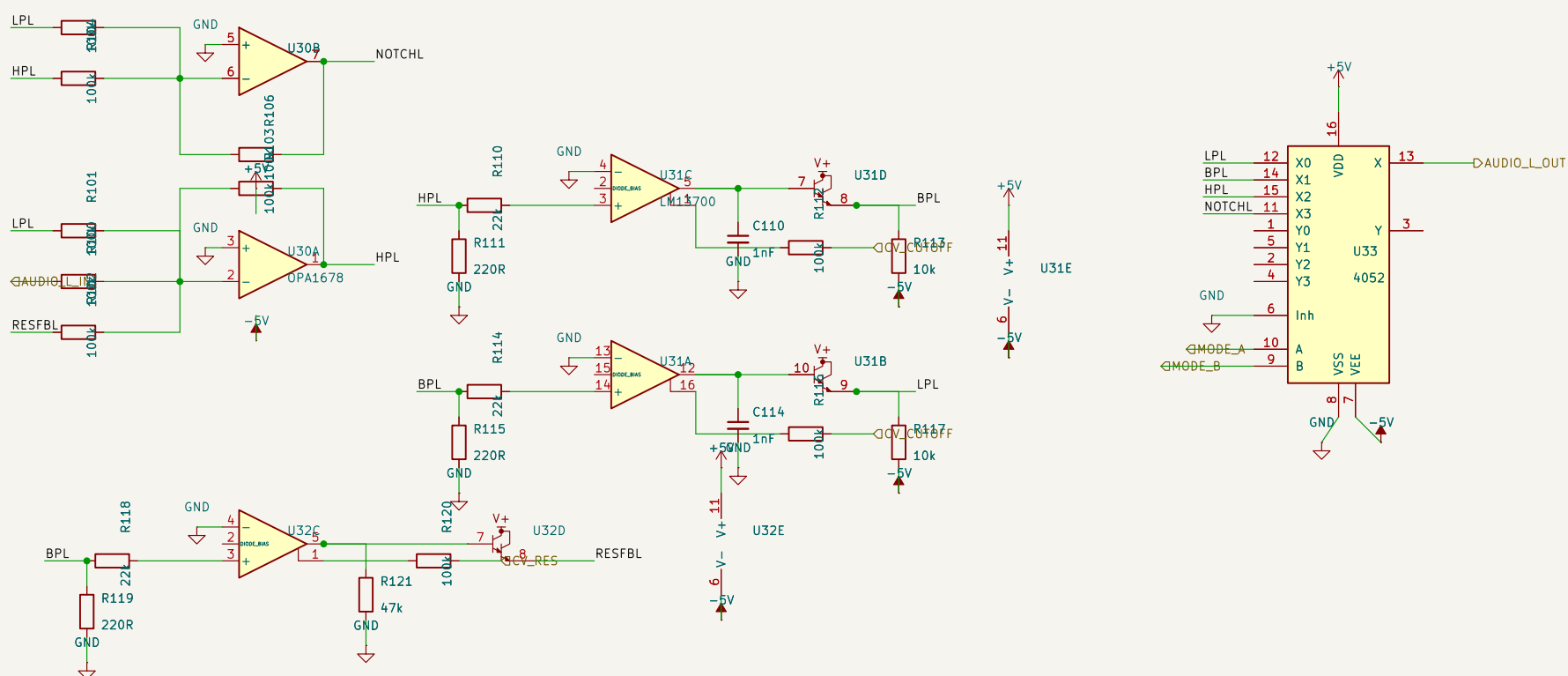
VCF+VCA R - LM13700 OTA lowpass then OTA VCA (NOTE: OTA/labc topology - verify vs datasheet)



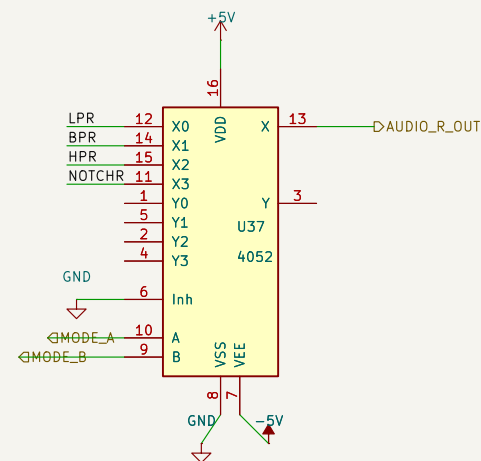
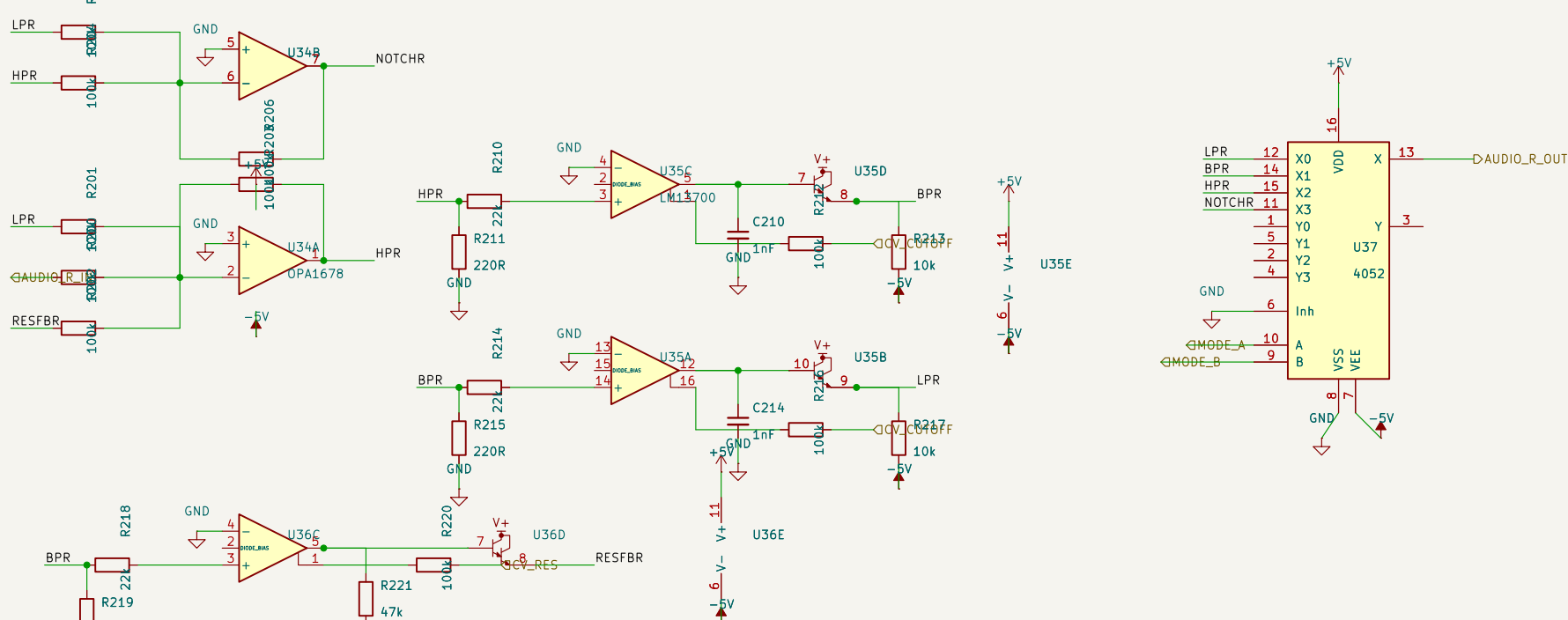
FILT_x_IN / FILT_x_OUT are the hierarchical pins shared with the root (front-end) sheet. Power rails (+5V/-5V/GND) are global.

Sheet: /ANALOG_FILTER_CV/ File: ANALOG_FILTER_CV.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 4/5

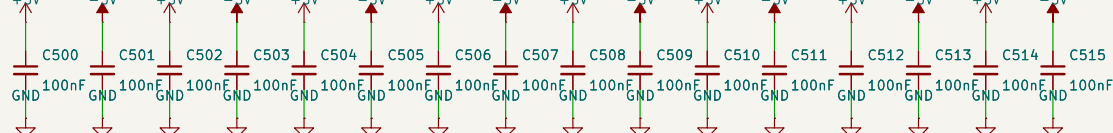
MASTER FILTER (stereo) - VC state-variable filter 2-pole: LP/HP/BP/notch; CV cutoff + CV resonance; mode-select mux



CHANNEL R



DECOUPLING – 100nF at each IC's supply pins (+5V and -5V)



NOTE: OTA VC - SVF¹ + OTA resonance VCA - prototype/verify levels, labc, buffer bias vs LM13700 datasheet before fab. ORDER_SEL reserved for the 2/4-pole cascade.

Sheet: /MASTER_FILTER/		
File: MASTER_FILTER.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 5/5